Lesson 1: Spatial Puzzles: Tangrams and Pentominoes

Assignment

1. Which strategy from Section 1 fits each plan best? Choose from: *draw or model, guess and check, systematic list, look for a pattern, simplify the problem, eliminate possibilities.*

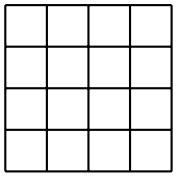
(a) Jess cuts a net out of paper and folds it to see if it makes a box. _____

(b) Raj lists every way to make \$1 with quarters and dimes, in order. _____

(c) Before a 10-piece puzzle, Mei solves a 3-piece version of it. _____

(d) For a 9-unit shape, Ty crosses off every set of pieces whose areas do not add to 9. _____

2. How many squares of any size are in a 4 by 4 grid?



(a) Complete the systematic list.

Size	1 by 1	2 by 2	3 by 3	4 by 4	**Total**
How many					

(b) Check your total with the pattern from Example 1.

3. Use the tangram pieces (small triangle = 1 unit of area) to answer.

(a) How many small triangles cover the parallelogram? _____

(b) How many medium triangles cover one large triangle? _____

(c) Which pieces have the same area as the square? _____

(d) What fraction of the whole tangram are the 5 smaller pieces together (all but the two large triangles)? _____

(e) Ava says, "There are 7 pieces, so each piece is $\frac{1}{7}$ of the tangram." Identify and correct her error.

4. A large tangram for a classroom wall is cut from a square of foam board 30 cm by 30 cm. Find the area of each kind of piece.

Whole square	**Small triangle**	**Medium triangle, square, parallelogram**	**Large triangle**

5. Use the **two small triangles and the square**. Make each shape and sketch how the pieces fit.

(a) a rectangle

(b) a triangle

(c) a trapezoid (one pair of parallel sides)

6. Make this rectangle with **all 7** tangram pieces. (The picture is smaller than your pieces.) Draw the lines, then describe your strategy.



7. Jordan wants to fill a silhouette with an area of 10 units (small triangle = 1 unit). Use area to eliminate the sets of pieces that cannot work.

Set of pieces	**Total area**	**Could it work?**
(a) both large triangles and the medium triangle		
(b) one large triangle, the square and both small triangles		
(c) the medium triangle, square, parallelogram and both small triangles		
(d) one large triangle, square, parallelogram and both small triangles		

8. Is each shape a pentomino? Write yes or no. If no, say why.

(a) (b) (c) (d) (e) (f)



_____ _____ _____ _____ _____ _____

9. Sort these six shapes into three pairs that are the same pentomino. For each pair, write its letter name.

1 2 3 4 5 6

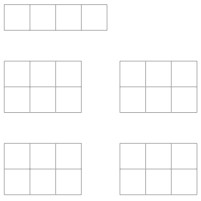


10. A **tetromino** is made of 4 squares joined along full sides. Find all of them with a systematic list sorted by the longest row. Shade one on each grid. How many are there?

4 in a row

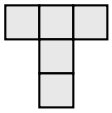
3 in a row

2 in a row

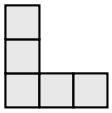


There are _____ tetrominoes.

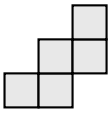
11. Does each pentomino fold into an open box? Write yes or no. If yes, shade the square that becomes the bottom. If no, explain.

(a) T 

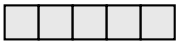
_____

(b) V 

_____

(c) W 

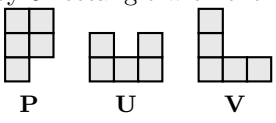
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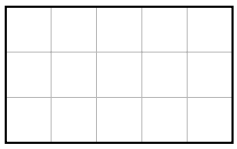
(d) I 

_____

12. Lee says, "Every pentomino with 4 squares in a row folds into an open box." Identify and correct Lee's error. Use a pentomino as an example.

13. Tile this 3 by 5 rectangle with the **P**, **U** and **V** pentominoes. Label each piece. Then describe one strategy you used.



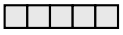


14. Create a variation of a tangram puzzle. Choose exactly 3 tangram pieces, put them together into a new shape, and sketch its outline (its silhouette). Give its area in small-triangle units, and write one tip that would help a partner solve it.

15. A tangram is cut from a square with an area of 64 cm^2. What is the area of the parallelogram?

A. 4 cm^2 **B.** 8 cm^2 **C.** 16 cm^2 **D.** 32 cm^2

16. Which pentomino folds into an open box?

A.  **B.**  **C.**  **D.** 

17. Numerical Response Kenji cuts a tangram from a square of card 12 cm by 12 cm. The area of the medium triangle is _____ cm^2.

18. Numerical Response Use the pattern from Example 1 and Question 2 $(1, 5, 14, 30, \dots)$. The number of squares of any size in a 5 by 5 grid is _____.

Lesson 2: Spatial Games and Strategies

Assignment

1. Match each description to a strategy from the toolbox.

Description	**Strategy**
(a) You try 1 disc, then 2 discs, then 3 discs before the 10-disc tower.	
(b) You start from the last move of a game and figure out the move before it.	
(c) You write down every possible first move, one at a time, in an order.	
(d) You notice the answers go 1, 3, 7, 15 and predict the next one.	
(e) You cross out ship positions that use a square you missed.	
(f) You use pennies on a table to act out the game.	

2. In the take-away game (take 1, 2 or 3; last counter wins), how many counters should you take from each pile to leave a multiple of 4? If you cannot, write "stuck".

Counters on your turn	9	18	19	30	28
Take					

3. A new take-away game: on your turn you take **1 or 2** counters. The player who takes the last counter wins.

(a) Work backward. Complete the table.

Counters on your turn	1	2	3	4	5	6	7	8	9
Win or lose?									

(b) What numbers should you leave for the other player?

(c) The pile has 13 counters and you go first. How many should you take?

4. In another version, you may take 1, 2, 3, 4 or 5 counters, and the last counter wins. Ravi's strategy is "leave a multiple of 5."

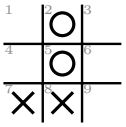
(a) Show that Ravi's strategy fails when he leaves 5 counters.

(b) What should the strategy be? Explain.

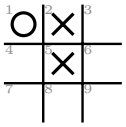
(c) With 20 counters, how many should Ravi take first?

5. Use the tic-tac-toe checklist. Where should the player whose turn it is play? Give the square number and the reason.

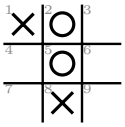
(a) X to play.



(b) O to play.



(c) X to play.



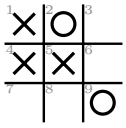
6. For the first move in tic-tac-toe there are 9 squares, but some are really the same move. If you turn the board a quarter turn, corner 1 becomes another corner.

(a) Sort the 9 squares into groups of moves that are really the same. How many really different first moves are there?

(b) Which strategy from the toolbox did you use? How does it make the game easier to study?

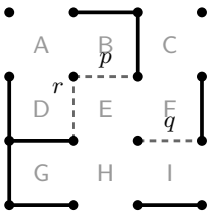
7. Here is a game between Ella (X) and Noor (O): X plays 1, O plays 2, X plays 5, O plays 9, X plays 4. Ella wins on her next turn.

(a) Here is the board after X plays 4. Why can't Noor stop Ella now?



(b) Noor's 9 was a forced block. Which earlier move was her mistake? Where should she have played?

8. It is your turn in dots and boxes. The dashed lines p, q and r have not been drawn.



(a) Which box can you take? Which line do you draw?

(b) You get another turn. Is p, q or r the safe line? Explain each.

9. Near the end of a dots-and-boxes game, every line left gives boxes away. Line p gives away a group of 1 box. Line q gives away a row of 4 boxes (once the first is taken, the next one has 3 sides, and so on).

(a) Which line should you draw?

(b) Arun says, "It doesn't matter which one. Either way I give away boxes." Correct him.

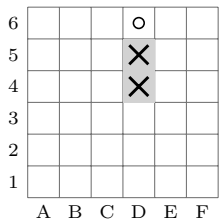
10. Answer these about the tower puzzle (3 pegs, A, B and C).

(a) List the fewest moves to take 2 discs from peg A to peg **B** (S = small, L = large).

(b) A puzzle has 10 discs. At 1 move per second with no mistakes, how long does it take? Give minutes and seconds.

(c) Owen says, "5 discs take 31 moves, so 6 discs take $2 \times 31 = 62$ moves." Find his error and correct it.

11. In the hidden-ships game you know the other player has a **4-ship** (4 squares in a line). You hit it at D4 and D5. Your call at D6 was a miss.



(a) Two hits are side by side in column D. Why must the ship run up and down?

(b) List every up-and-down position for the 4-ship that covers D4 and D5. Eliminate the ones that cannot work.

(c) Which squares will sink the ship?

12. You hunt for ships on an 8×8 grid using a checkerboard pattern.

(a) How many shaded squares do you need to call (at most) to find every ship?

(b) Explain why a 3-ship can never hide only on the white squares.

(c) **Variation:** the other player adds a 1-square ship. Does the checkerboard hunt still find it for sure? Explain.

13. Make your own variation. Choose the take-away game, tic-tac-toe or dots and boxes. Change **one** rule. Write the new rule, play it twice with someone (or against yourself), and describe a strategy for winning the new game.

14. In the take-away game (take 1, 2 or 3; last counter wins), the pile has 26 counters and you go first. How many should you take?

A. 1 **B.** 2 **C.** 3 **D.** It doesn't matter.

15. What is the fewest number of moves for a tower puzzle with 5 discs?

A. 10 **B.** 25 **C.** 31 **D.** 32

16. Which tic-tac-toe move is a **fork**?

A. A move that makes two in a row in two lines at once

B. A move that blocks the other player's two in a row

C. Taking the centre on the first move

D. A move that completes three in a row

17. Numerical Response The fewest number of moves for a tower puzzle with 7 discs is _____.

18. Numerical Response In a take-away game you may take 1, 2, 3 or 4 counters, and the last counter wins. The pile has 38 counters and you go first. To be sure of winning, you should take _____ counters.

Lesson 3: Measuring, Drawing and Classifying Angles

Assignment

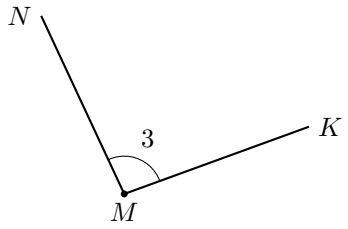
1. Use the picture.

(a) Which point is the vertex? _____

(b) Name the two arms. _____

(c) Name the angle in three ways.

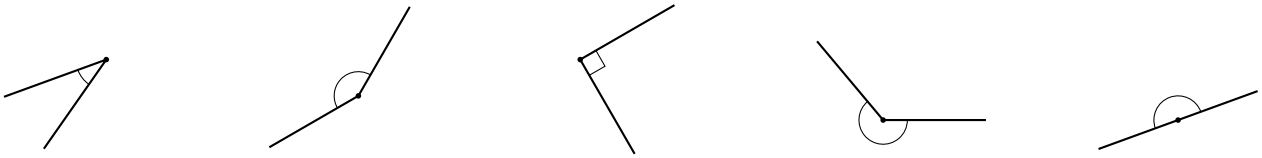
_____



2. Write the type of each angle: acute, right, obtuse, straight or reflex.

Angle	23°	178°	90°	241°	95°	359°	180°	60°
Type								

3. Write the type of each marked angle.



(a) _____ (b) _____ (c) _____ (d) _____ (e) _____

4. What type of angle is each one? Explain how you know.

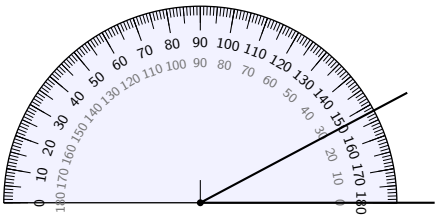
(a) The angle between a wall and the floor.

(b) The point of a slice of pie, when the pie is cut into 6 equal slices.

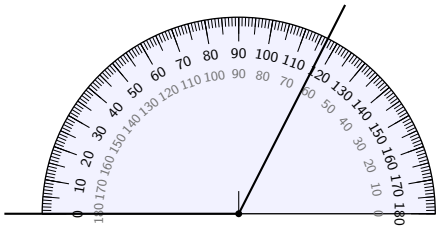
(c) The angle between a car seat and its seat back, when the seat back is leaning back a little.

(d) The angle between a skateboard ramp and the ground it sits on.

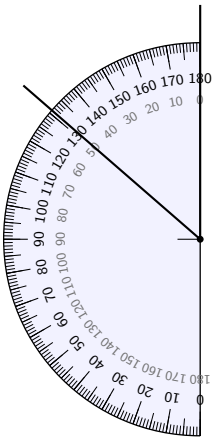
5. Read each protractor. Give the size of the angle.



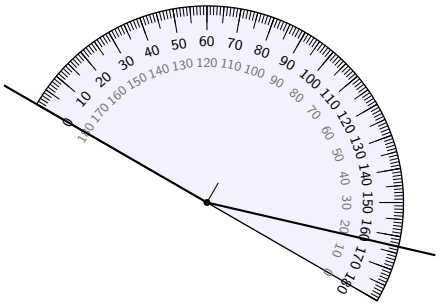
(a) _____



(b) _____



(c) _____

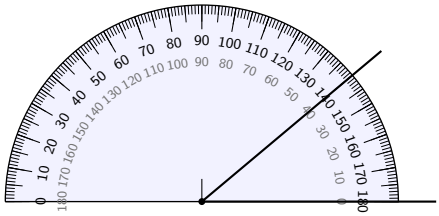


(d) _____

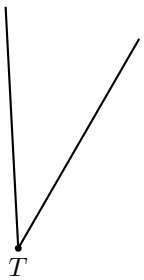
6. Mai measured this angle and wrote 140°.

(a) What mistake did she make?

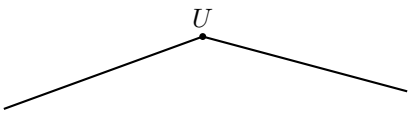
(b) What is the correct measure? How can she check?



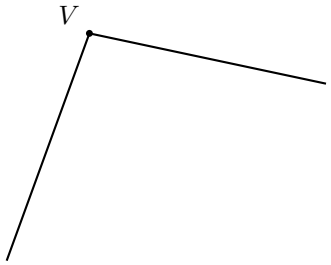
7. Use your protractor to measure each angle. (The pictures are full size.) Then write its type.



(a) $\angle T =$ _____ _____



(b) $\angle U =$ _____ _____



(c) $\angle V =$ _____ _____

8. Use a protractor and a ruler to draw each angle on the arm given. Mark and label the angle.

(a) 40°



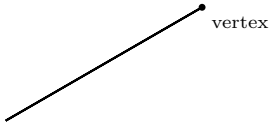
(b) 125°



(c) 90°



(d) 165°



9. Find the reflex angle that goes with each smaller angle in (a) to (d).

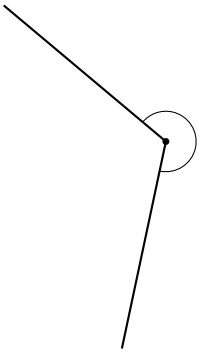
(a) 30°

(b) 145°

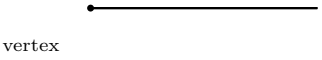
(c) 90°

(d) 179°

(e) Measure the marked reflex angle.



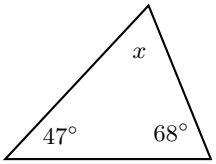
(f) Draw a reflex angle of 300° on the arm given.



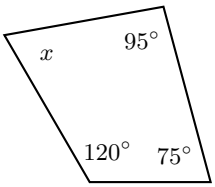
10. Complete the table. Round to the nearest tenth of a degree if needed.

Polygon	**Sides, n**	**Sum of the angles**	**Each angle, if regular**
(a) pentagon	5		
(b) heptagon			
(c) decagon			
(d) dodecagon			

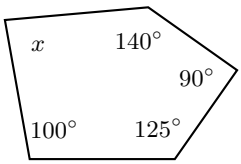
11. Find the missing angle x in each shape.



(a)



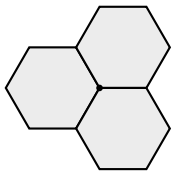
(b)



(c)

12. A landscaper in Airdrie is building a patio from paving stones shaped like regular hexagons.

(a) Find the size of each angle of one paving stone.



(b) Three stones meet at each point, as in the picture. Explain why they fit with no gaps.

13. Kai says the interior angles of a polygon with 13 sides add up to 2340°. Find his mistake and give the correct sum.

14. Each interior angle of a regular polygon is 156°. How many sides does it have? (Try a few values of n and make a table, or work backward.)

15. Which angle is obtuse?

A. 89° **B.** 90° **C.** 135° **D.** 210°

16. The smaller angle between two arms is 65°. The reflex angle is

A. 25° **B.** 115° **C.** 245° **D.** 295°

17. Numerical Response A regular polygon has 20 sides. Each of its interior angles is _____°.

18. Numerical Response The sum of the interior angles of a polygon with 17 sides is _____°.

Lesson 4: Estimating Angles with Referents, and True Bearings

Assignment

1. Find the smaller angle between the hands of a clock at each time.

(a) 9:00 _____ (b) 10:00 _____ (c) 7:00 _____ (d) 11:00 _____

2. Rosa folds a square sheet of paper to make referent angles. Fill in the blanks.

(a) The corner of the sheet: _____ (b) Fold the corner in half: _____ (c) Fold that in half again: _____

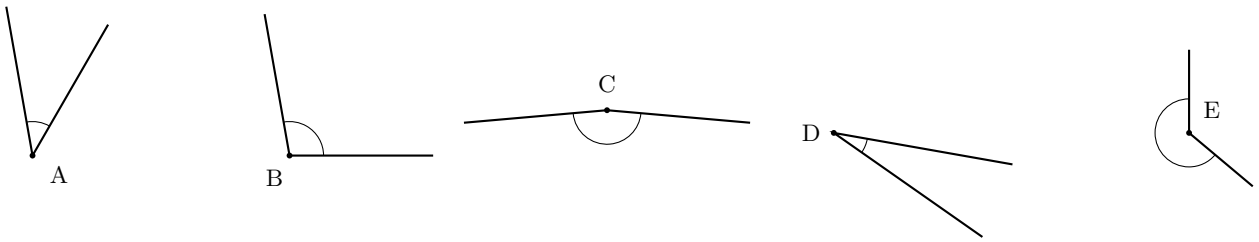
(d) She folds a straight edge of the paper onto itself, so the fold crosses the edge. Each angle at the fold is _____, because _____.

3. Build each angle from referents. Then name its type.

(a) 75° (b) 105° (c) 210°

(d) 300° (e) 157.5° (f) 120°

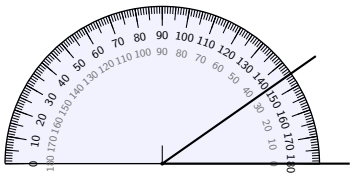
4. Estimate each angle using referents. Then measure it with a protractor.



Angle	A	B	C	D	E
Estimate					
Measure					

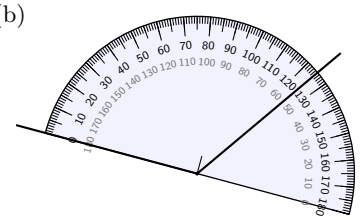
5. Estimate each angle. Then read the protractor and write the angle.

(a)



Estimate: _____ Angle: _____

(b)



Estimate: _____ Angle: _____

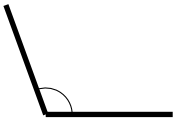
6. Jordan measures an angle that is clearly a little less than a right angle. He writes 100°.

(a) How do you know his answer is wrong without measuring?

(b) What mistake did he likely make, and what is the correct angle?

7. Estimate the marked angle in each picture.

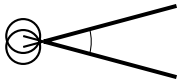
(a) An open laptop



(b) A skateboard ramp



(c) Open scissors



8. Sketch each angle without a protractor. Write the referents you used. Then check with a protractor.

(a) 30° (b) 135° (c) 100° (d) 290°

9. Complete the table.

Direction	N		SE		NW	SSW	
True bearing		090°		270°			157.5°

10. Write each direction as a true bearing. Make a quick sketch for each.

(a) 35° east of north (b) 10° south of east (c) 50° west of south

(d) 20° north of west (e) 25° west of north (f) 60° east of south

11. Draw a line from each point P in the direction of the bearing.

(a) 

040°

(b) 

110°

(c) 

200°

(d) 

290°

12. An orienteering course starts at S and visits checkpoints 1, 2 and 3. Use a protractor.

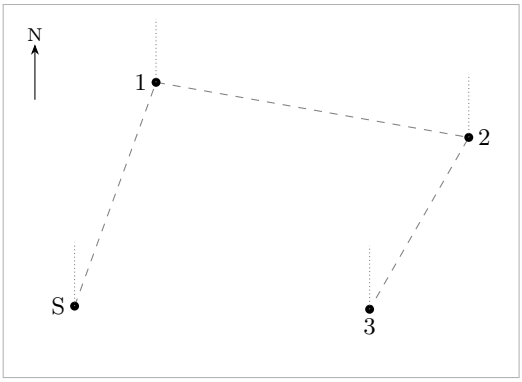
(a) Bearing from S to checkpoint 1 _____

(b) Bearing from 1 to 2 _____

(c) Bearing from 2 to 3 _____

(d) Bearing from 3 to S _____

(e) A runner at checkpoint 2 wants to go back to checkpoint 1. On what bearing should she run? Use the rule, not the protractor.



13. Find the bearing back for each bearing.

Bearing	050°	165°	240°	300°	010°
Bearing back					

14. Mila paddles her canoe from the dock at Sylvan Lake to an island on a bearing of 125°. Later her friend Tyrell says, "To get back, paddle on $125° - 180°$, so on $-55°$."

(a) What is wrong with Tyrell's answer?

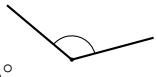
(b) On what bearing should Mila paddle back to the dock?

15. A surveyor stands at a corner of a city lot in Airdrie. She sights a power pole on a bearing of 040° and a fire hydrant on a bearing of 115°. What is the angle between her two sightings? Is it acute or obtuse?

16. A boat is heading southeast. What is its true bearing?

A. 045° **B.** 135° **C.** 225° **D.** 315°

17. Which is the best estimate of this angle?



A. 35° **B.** 125° **C.** 175° **D.** 235°

18. Numerical Response A bush plane flies from Yellowknife to a mining camp on a bearing of 312°. The bearing it must fly to return to Yellowknife is _____°.

19. Numerical Response A boat is heading NNE. The captain turns the boat clockwise until it is heading due east. The angle the boat turns, in degrees, is _____.

Lesson 5: Copying and Bisecting Angles; Perpendicular Lines

Assignment

1. Fill in each blank with a word from the list: **bisect, perpendicular, parallel, congruent, midpoint**.

(a) Two angles with the same size are _____.

(b) Lines that meet at 90° are _____.

(c) To cut an angle into two equal angles is to _____ it.

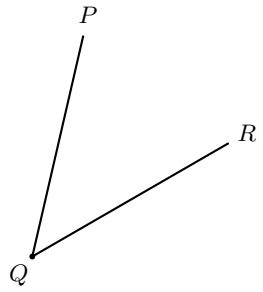
(d) Lines on a flat surface that never meet are _____.

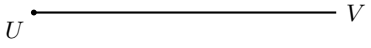
(e) The point that cuts a segment into two equal parts is its _____.

2. Each angle is bisected. Complete the table.

	(a)	(b)	(c)	(d)	(e)	(f)	(g)
Whole angle	60°	110°	45°	174°	33°		
Each half						19°	67.5°

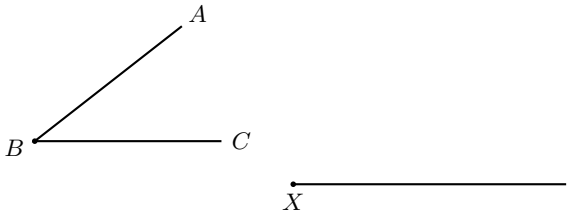
3. Measure $\angle PQR$ with a protractor. Then use the protractor to copy it onto ray UV, with its vertex at U.



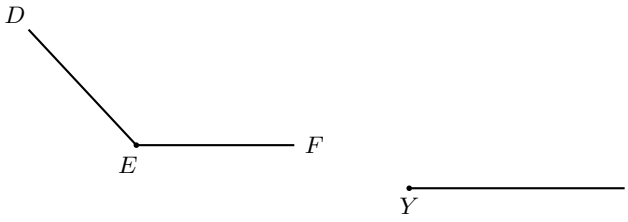


4. Copy each angle onto the ray beside it using only a compass and a straightedge. Then measure both angles to check.

(a)

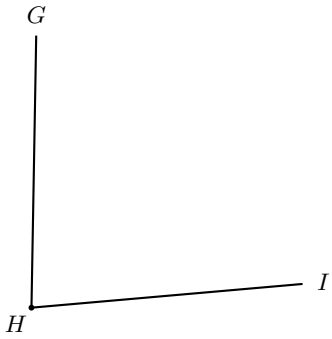


(b)

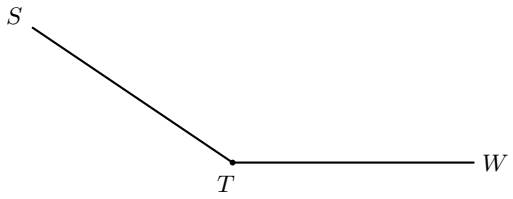


5. Bisect each angle with a compass and straightedge. Then measure each half to check. What should each half be?

(a)



(b)



6. Take a square sheet of paper (or a sticky note).

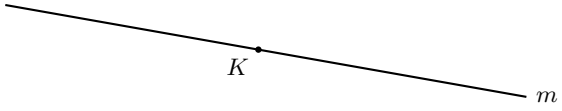
(a) Fold one corner so that its two edges lie on top of each other. The crease bisects the corner. What is each angle at the corner now?

(b) Fold again to bisect one of those angles. What size is the new angle?

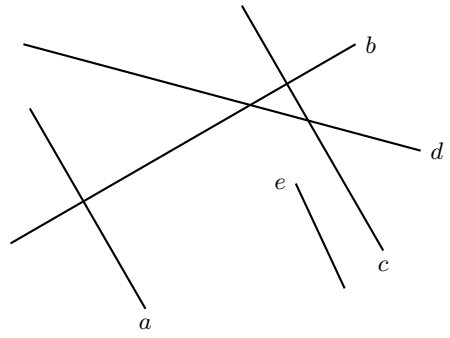
7. Construct the perpendicular bisector of segment CD. Label the midpoint M. Measure CM and MD, and the angle at M.



8. Construct a line through K that is perpendicular to line m. Check the angle with a protractor or the corner of a sheet of paper.



9. Use a ruler and a protractor. Sort each pair of lines as parallel, perpendicular or neither, and give a reason.



Pair	Sort	Reason
a, c		
a, b		
b, c		
b, d		
d, e		
c, d		

10. Are these lines parallel, perpendicular or neither?

(a) the two rails of a straight railway track _____

(b) the hands of a clock at 3:00 _____

(c) the hands of a clock at 1:00 _____

(d) the side edge of a door and the floor _____

(e) two lines between angled parking stalls _____

(f) an angled parking line and the curb _____

11. A fence post makes an angle of 83.5° with the ground on its left side.

(a) What angle does it make with the ground on its right side?

(b) By how many degrees must the post be straightened to be perpendicular to the ground?

(c) Kiran checks the post with a carpenter's square from the front only. Is that enough? Explain.

12. A framer is making picture frames.

(a) What is the cut angle at each end of a piece for a square frame?

(b) A frame is a regular octagon (8 sides), like a stop sign. Find the corner angle, the cut angle and the mitre saw setting.

13. Leah copies an angle with a compass. Here is what she did:

Step 1: arc from the vertex B, crossing both arms at P and Q.
Step 2: she opened the compass *wider*, then drew a long arc from the new vertex E, crossing the ray at R.
Step 3: arc of width QP from R, crossing the long arc at S. Step 4: ray ES.

Her copy does not match the angle. Find her mistake and explain how to fix it.

14. Puzzles. Use guess and check if it helps.

(a) An angle is bisected. Then one of the halves is bisected again. The smallest angle is 17°. What was the first angle?

(b) An angle and one of its halves add up to 90°. What is the angle?

15. An angle of 136° is bisected. What is the size of each new angle?

A. 34° **B.** 44° **C.** 68° **D.** 272°

16. Which tool can copy a right angle but **cannot** copy a 72° angle?

A. a protractor

B. a compass and straightedge

C. a carpenter's (framing) square

D. a sliding T-bevel

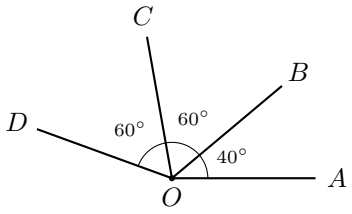
17. Numerical Response A frame is shaped like a regular pentagon (5 sides). The cut angle at each end of a piece, in degrees, is _____°.

18. Numerical Response An angle is bisected. Each half measures 71.5°. The size of the original angle, in degrees, is _____°.

Lesson 6: Angle Partners: Complementary, Supplementary and Opposite Angles

Assignment

1. Use the diagram. Rays OA, OB, OC and OD start at point O.



Are the angles adjacent? Write yes or no, and give a reason for each no.

(a) $\angle AOB$ and $\angle BOC$ _____

(b) $\angle AOB$ and $\angle COD$ _____

(c) $\angle AOC$ and $\angle BOC$ _____

(d) $\angle BOC$ and $\angle COD$ _____

(e) Find $\angle AOC$. _____ (f) Find $\angle BOD$. _____

2. Complete the table. If an angle has no complement or no supplement, write "none" and say why.

Angle	Complement	Supplement
(a) 15°		
(b) 72°		
(c) 44°		
(d) 135°		
(e) 179°		
(f) 260°		

3. Work backwards.

(a) The complement of an angle is 67°. Find the angle and its supplement.

(b) The supplement of an angle is 101°. Find the angle and its complement.

(c) An angle and its complement are the same size. How big is the angle?

(d) An angle and its supplement are the same size. How big is the angle?

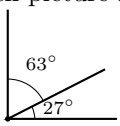
4. True or false? Explain each answer.

(a) Two acute angles can be supplementary.

(b) Two right angles are supplementary.

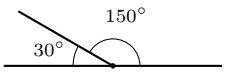
(c) An obtuse angle has a complement.

5. Each picture shows two adjacent angles. Find the sum. Then write *complementary*, *supplementary* or *neither*.

(a) 

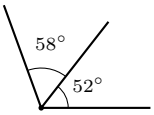
Sum: _____

_____

(b) 

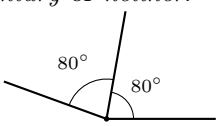
Sum: _____

_____

(c) 

Sum: _____

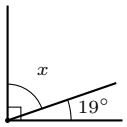
_____

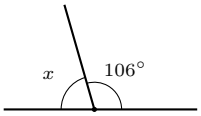
(d) 

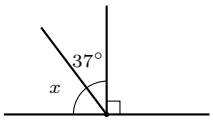
Sum: _____

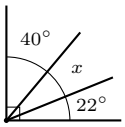
_____

6. Find each missing angle x. Write the equation first.

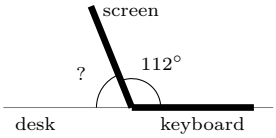
(a) 

(b) 

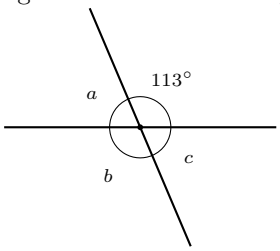
(c) 

(d) 

7. Ella's laptop sits on a flat desk. She opens the screen so that it makes a 112° angle with the keyboard. What angle does the back of the screen make with the desk behind it? Explain.

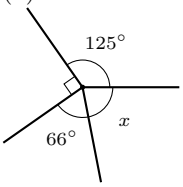


8. Two straight lines cross. Find a, b and c. Give a reason for each.



9. Angles around a point.

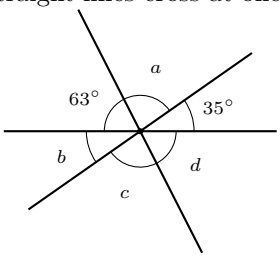
(a) Find x.



(b) A Ferris wheel has 12 cars, evenly spaced around the centre. What angle is between two cars that are next to each other?

(c) An angle measures 75°. How big is the reflex angle that goes the long way around?

10. Three straight lines cross at one point. Find a, b, c and d. Give a reason for each.



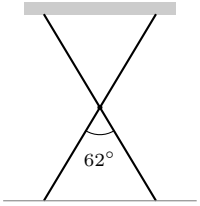
11. Two straight lines cross, and one of the angles is 61°. Mia says, "The angle across from the 61° angle is 119°, because they are supplementary."

(a) What mistake did Mia make?

(b) Give the correct sizes of the other three angles.

12. A hockey player passes the puck off the straight boards to a teammate. The puck hits the boards at a 32° angle and comes off at a 32° angle (on the other side). What is the angle p between the puck's path in and its path out?

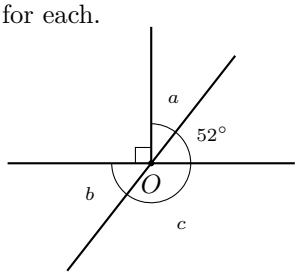
13. The legs of a folding camp stool cross like an X. The angle between the legs at the floor end is 62°.



(a) What is the angle between the legs at the top (under the seat)? Why?

(b) What are the angles on the left and right sides of the X? Why?

14. Two straight lines cross at O. A third line from O makes a right angle with the flat line. Find a, b and c. Give a reason for each.



15. What is the complement of 58°?

A. 122° **B.** 32° **C.** 42° **D.** 302°

16. Two straight lines cross. One of the four angles is 47°. What are the other three angles?

A. 133°, 47°, 133°

B. 43°, 47°, 43°

C. 47°, 47°, 47°

D. 133°, 133°, 133°

17. Which pair of adjacent angles is **neither** complementary nor supplementary?

A. 37° and 53°

B. 112° and 68°

C. 45° and 75°

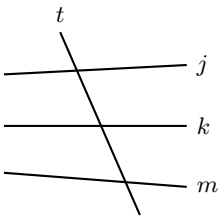
D. 90° and 90°

18. Numerical Response Four angles meet at a point with no gaps. Three of them are 104°, 87° and 95°. The size of the fourth angle, in degrees, is _____.

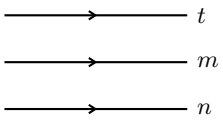
Lesson 7: Transversals: Naming the Angle Pairs

Assignment

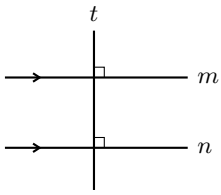
1. Is line t a transversal of the other lines? Write yes or no, and give a reason.



(a) _____

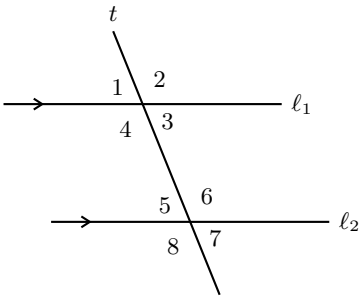


(b) _____



(c) _____

2. Questions 2 to 4 use this diagram. The angles are numbered the usual way: 1 to 4 clockwise from the upper left at the top crossing, 5 to 8 clockwise from the upper left at the bottom crossing.



(a) Which lines are parallel? How do you know?

_____

(b) List the interior angles. _____

(c) List the exterior angles. _____

(d) List the angles on the left side of t. _____

(e) Which interior angles are on the right side of t? _____

3. Use the diagram in Question 2. Name the angle that is:

(a) corresponding to $\angle 7$ _____ (b) corresponding to $\angle 1$ _____

(c) vertically opposite $\angle 6$ _____ (d) vertically opposite $\angle 3$ _____

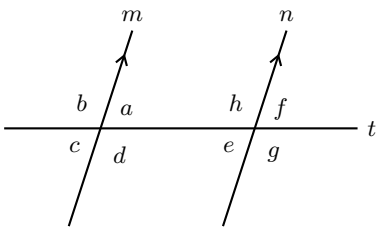
(e) alternate interior to $\angle 3$ _____ (f) alternate exterior to $\angle 8$ _____

(g) interior on the same side as $\angle 6$ _____ (h) exterior on the same side as $\angle 1$ _____

4. Use the diagram in Question 2. Name each pair of angles. If they are at the same crossing, say whether they are vertically opposite or make a straight line. If the pair has no special name, say so.

Pair	Name	Pair	Name
(a) $\angle 4$ & $\angle 8$		(e) $\angle 5$ & $\angle 7$	
(b) $\angle 2$ & $\angle 7$		(f) $\angle 5$ & $\angle 6$	
(c) $\angle 4$ & $\angle 6$		(g) $\angle 3$ & $\angle 8$	
(d) $\angle 1$ & $\angle 7$			

5. In this diagram the transversal t goes across the page, and the two parallel lines m and n go up and down. The angles are labelled with letters.



Name each pair.

(a) $\angle a$ and $\angle f$ _____

(b) $\angle d$ and $\angle h$ _____

(c) $\angle a$ and $\angle h$ _____

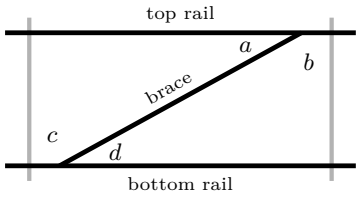
(d) $\angle c$ and $\angle f$ _____

(e) $\angle c$ and $\angle e$ _____

(f) $\angle b$ and $\angle f$ _____

6. Use the diagram in Question 2. Mia says that $\angle 2$ and $\angle 6$ are alternate exterior angles "because they are at different crossings". Is she right? Explain.

7. A farm gate has a top rail and a bottom rail that are parallel. A diagonal brace crosses between them.

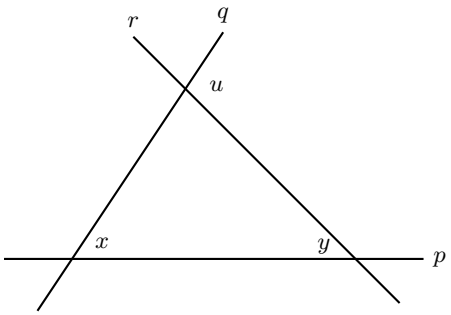


(a) For these four angles, what are the two lines and what is the transversal?

(b) Name the pair $\angle a$ and $\angle d$. What letter shape do they make?

(c) Name the pair $\angle a$ and $\angle c$. (d) Name the pair $\angle b$ and $\angle c$.

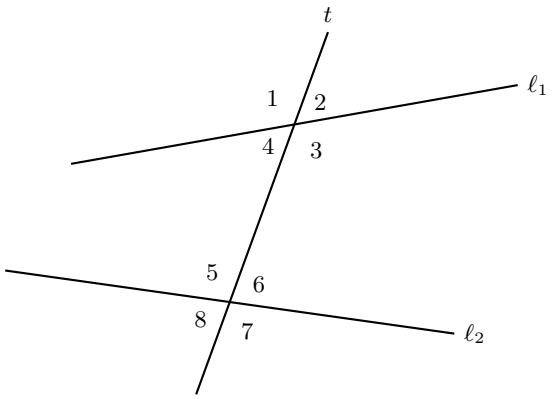
8. Three lines p, q and r cross each other and make a triangle.



(a) For $\angle x$ and $\angle y$, which line is the transversal, and which are the two lines? Name the pair.

(b) For $\angle x$ and $\angle u$, which line is the transversal, and which are the two lines? Name the pair.

9. In this diagram, ℓ_1 and ℓ_2 are not parallel. The angles are numbered the usual way.



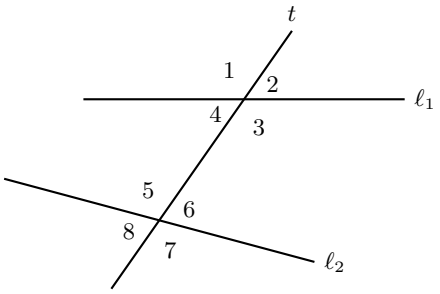
(a) Measure with a protractor.

$\angle 3 =$ _____ $\angle 7 =$ _____

$\angle 4 =$ _____ $\angle 6 =$ _____

(b) Name the pairs $\angle 3$ & $\angle 7$, and $\angle 4$ & $\angle 6$. Are the angles in each pair equal? Why or why not?

10. In this diagram, ℓ_1 and ℓ_2 are **not** parallel. $\angle 1 = 125°$ and $\angle 6 = 70°$.



(a) Find the other six angles. Give a reason for each.

(b) Are the alternate interior angles $\angle 3$ and $\angle 5$ equal? Do the interior angles on the same side, $\angle 3$ and $\angle 6$, add to 180°?

11. A diagram has two lines and a transversal, numbered the usual way, with **no arrow marks**. $\angle 4 = 65°$. Jordan says, "$\angle 6 = 65°$ because alternate interior angles are equal."

(a) Is Jordan right? Explain.

(b) Which other angle can you be *sure* is 65°? Why?

12. Use your answers to Question 10. Add the four interior angles: $\angle 3 + \angle 4 + \angle 5 + \angle 6$. Then explain why the four interior angles add to the same total in *every* diagram, even when the lines are not parallel.

13. In the usual numbering, which pair are alternate exterior angles?

A. $\angle 1$ and $\angle 5$ **B.** $\angle 2$ and $\angle 8$ **C.** $\angle 4$ and $\angle 6$ **D.** $\angle 2$ and $\angle 4$

14. In the usual numbering, $\angle 4$ and $\angle 5$ are

A. alternate interior **B.** corresponding **C.** interior on the same side **D.** alternate exterior

15. Two lines are cut by a transversal. Which statement is true **whether or not** the lines are parallel?

A. Corresponding angles are equal.

B. Alternate interior angles are equal.

C. Interior angles on the same side add to $180°$.

D. Vertically opposite angles are equal.

16. Numerical Response Two lines are cut by a transversal, numbered the usual way. The lines are **not** parallel. $\angle 7 = 116°$ and $\angle 2 = 58°$. The value of $\angle 5 + \angle 4$, in degrees, is _____.

Lesson 8: Parallel Lines and Transversals: Finding Missing Angles

Assignment

1. Fill in the blanks. When two parallel lines are cut by a transversal:

(a) corresponding angles are _____.

(b) alternate interior angles are _____.

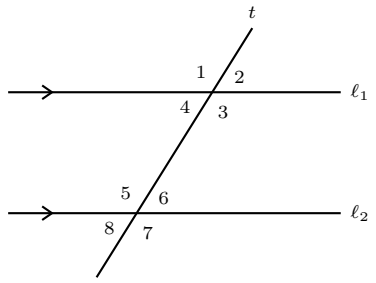
(c) alternate exterior angles are _____.

(d) interior angles on the same side add to _____.

(e) exterior angles on the same side add to _____.

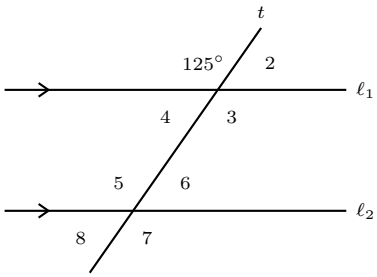
(f) vertically opposite angles are _____.

2. $\ell_1 \parallel \ell_2$. For each pair, write the name of the pair, and write **equal** or **add to 180°**.



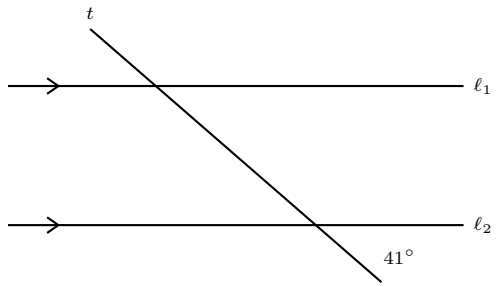
Pair	Name	Equal or add to 180°?
(a) $\angle 1$ and $\angle 5$		
(b) $\angle 4$ and $\angle 6$		
(c) $\angle 3$ and $\angle 6$		
(d) $\angle 2$ and $\angle 8$		
(e) $\angle 6$ and $\angle 8$		
(f) $\angle 1$ and $\angle 8$		
(g) $\angle 5$ and $\angle 6$		

3. $\ell_1 \parallel \ell_2$ and $\angle 1 = 125°$. Find each angle and give a reason.

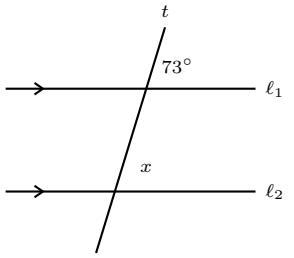


Angle	Size	Reason
(a) $\angle 5$		
(b) $\angle 3$		
(c) $\angle 7$		
(d) $\angle 8$		
(e) $\angle 6$		

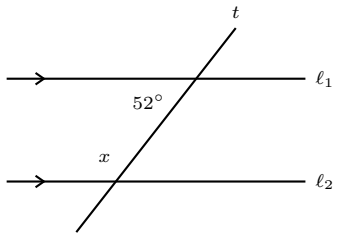
4. $\ell_1 \parallel \ell_2$ and $\angle 7 = 41°$. Write the size of every other angle in its corner of the diagram.



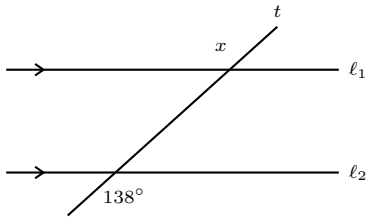
5. $\ell_1 \parallel \ell_2$. For each diagram, name the pair of angles, write an equation, and solve for x.



(a)

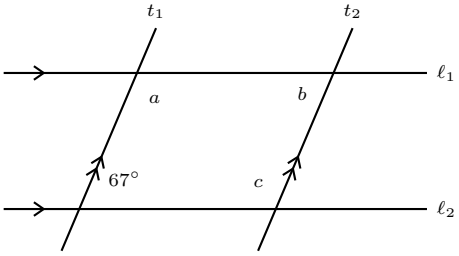


(b)



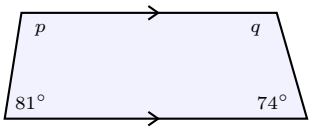
(c)

6. $\ell_1 \parallel \ell_2$ and $t_1 \parallel t_2$. Find a, b and c, in that order. Give a reason for each step.

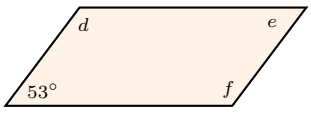


Angle	Size	Reason
a		
b		
c		

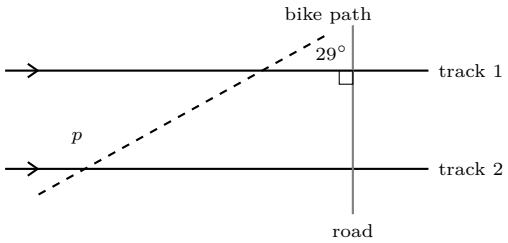
7. A window over a door is shaped like a trapezoid. Its top and bottom edges are parallel. The two bottom angles are 81° and 74°. Find the two top angles, p and q. Give your reason.



8. A floor tile is a parallelogram (both pairs of opposite sides are parallel). One of its angles is 53°. Find the other three angles, d, e and f.



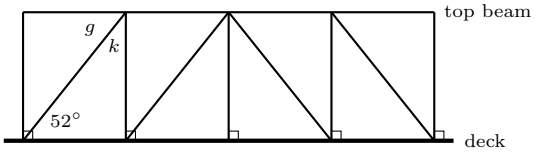
9. Two railway tracks are parallel. A road crosses both tracks, and it is perpendicular to track 1. A bike path also crosses both tracks. It makes a 29° angle with track 1.



(a) What angle does the road make with track 2? Why?

(b) Find angle p, where the bike path meets track 2. Show your steps.

10. In this bridge, the top beam is parallel to the deck. The posts are perpendicular to the deck. A diagonal brace makes a 52° angle with the deck.

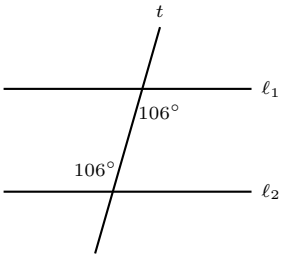


(a) What angle does each post make with the top beam? Why?

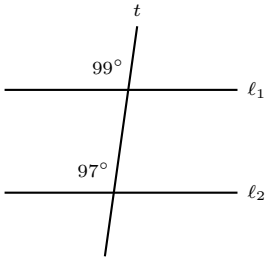
(b) Find g, the angle between the brace and the top beam. Give your reason.

(c) Find k, the angle between the brace and the post.

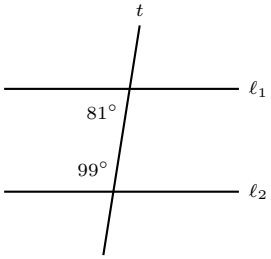
11. Are ℓ_1 and ℓ_2 parallel? Give the rule you used. (The diagrams are not drawn to scale.)



(a)

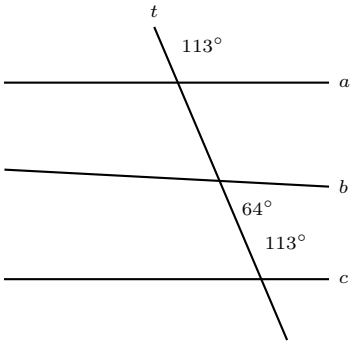


(b)



(c)

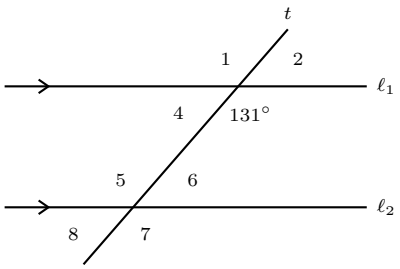
12. Three lines a, b and c are crossed by a transversal t. Which two lines are parallel? Explain. (Not drawn to scale.)



13. $\ell_1 \parallel \ell_2$ and $\angle 3 = 131°$. Ella writes:
"$\angle 6 = 131°$, because $\angle 3$ and $\angle 6$ are alternate interior angles, and those are equal."

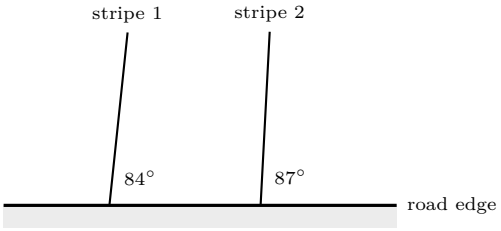
(a) What mistake did Ella make?

(b) Find the correct size of $\angle 6$.



14. The white stripes of a crosswalk must be parallel. The road edge is straight. A worker measures where two stripes meet the road edge: 84° for stripe 1 and 87° for stripe 2, both on the right side of the stripe, above the edge.

(a) Are the stripes parallel? Explain.



(b) Stripe 1 is correct. What angle should stripe 2 make with the road edge? By how many degrees must it change?

(c) Once stripe 2 is fixed, what angle does it make with the road edge on its *left* side?

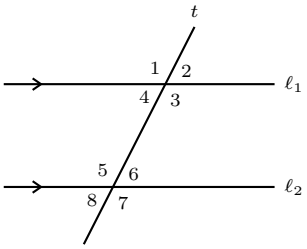
15. $\ell_1 \parallel \ell_2$. Which pair of angles adds to 180°?

A. $\angle 2$ and $\angle 6$

B. $\angle 3$ and $\angle 5$

C. $\angle 4$ and $\angle 5$

D. $\angle 1$ and $\angle 7$



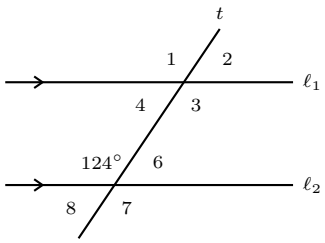
16. $\ell_1 \parallel \ell_2$ and $\angle 5 = 124°$. What is the size of $\angle 4$?

A. $124°$

B. $56°$

C. $236°$

D. $34°$



17. Two lines are cut by a transversal (angles numbered as in the diagrams above). Which measurements show that the lines **are** parallel?

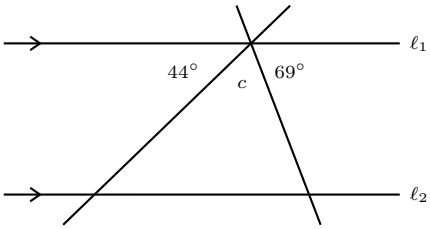
A. $\angle 1 = 104°$ and $\angle 3 = 104°$

B. $\angle 3 = 104°$ and $\angle 6 = 104°$

C. $\angle 2 = 76°$ and $\angle 6 = 76°$

D. $\angle 4 = 76°$ and $\angle 5 = 76°$

18. Numerical Response $\ell_1 \parallel \ell_2$. Two ramps meet at A on ℓ_1 and make a triangle with ℓ_2. The angles at A outside the triangle are $44°$ and $69°$. The size of the angle c inside the triangle at A is _____ °.



Lesson 9: Unit 5 Review

Assignment

This review is set up like the unit test. Show your work, and give a reason for every angle you find. A protractor, a compass and a ruler are allowed.

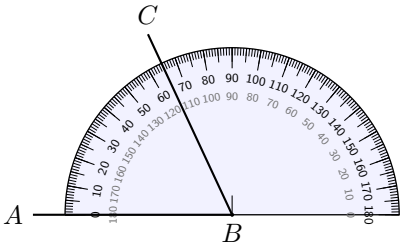
Part A: Written Response

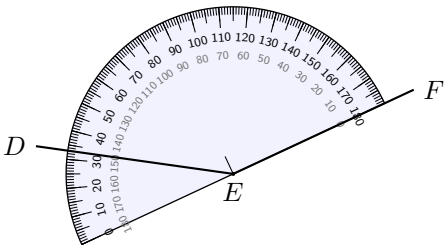
1. Classify each angle as acute, right, obtuse, straight or reflex.

(a) 37° _____ (b) 90° _____ (c) 154° _____

(d) 180° _____ (e) 263° _____ (f) 91° _____

2. Read each angle on the protractor. Then classify it.

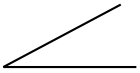




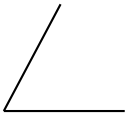
(a) $\angle ABC =$ _____ type: _____ (b) $\angle DEF =$ _____ type: _____

(c) Find the reflex angle that goes with $\angle DEF$.

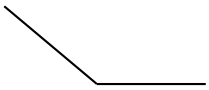
3. Estimate each angle using referents (22.5°, 30°, 45°, 60°, 90°, 180°). Do not measure.



(a) about _____



(b) about _____



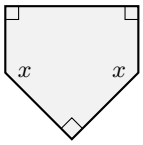
(c) about _____

(d) Sketch an angle of about 120° without a protractor. Which referents did you use?

4. Fill in the table. If there is no such angle, write "none".

Angle	Type	Complement	Supplement
19°			
81°			
126°			
			142°
		23°	

5. Home plate on a baseball field is a pentagon with three right angles and two equal angles, x.

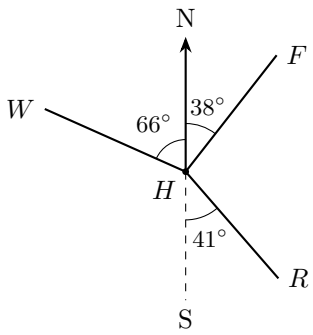


(a) Draw all the diagonals from one corner of the pentagon. How many triangles do you get? Use them to find the sum of the angles.

(b) Find the size of each angle x.

(c) A company wants to make floor tiles shaped like **regular** pentagons. Each angle is $540° \div 5 = 108°$. Can these tiles cover a floor with no gaps and no overlaps? Explain using the angles where tiles meet at a point.

6. A snowmobile club has a warming hut, H, near Grande Prairie. F is a fire tower, R is a river crossing and W is the west gate of the trail system.



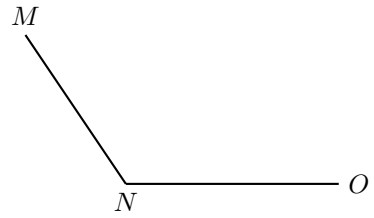
(a) Give the true bearing from the hut to F, to R and to W.

(b) A rider goes to the fire tower. On what bearing does she ride back to the hut?

(c) From the west gate, on what bearing do riders travel to the hut?

(d) One rider leaves the hut heading south-west. Write that as a true bearing.

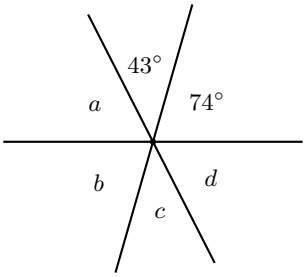
7. Bisect $\angle MNO$ with a compass and a straightedge. Leave your arcs showing.



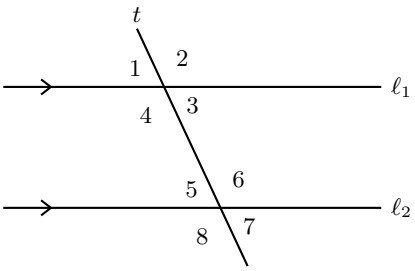
(a) Measure $\angle MNO$ and each half after you bisect it.

(b) Describe another way to bisect an angle drawn on paper.

8. Find a, b, c and d. Give a reason for each.



9. Use the diagram to name each pair of angles.



(a) $\angle 4$ and $\angle 6$ _____

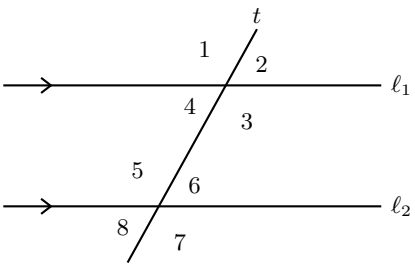
(b) $\angle 3$ and $\angle 6$ _____

(c) $\angle 1$ and $\angle 5$ _____

(d) $\angle 2$ and $\angle 7$ _____

(e) $\angle 5$ and $\angle 7$ _____

10. In the diagram, $\ell_1 \parallel \ell_2$ and $\angle 4 = 61°$.



Find each angle and give a reason.

(a) $\angle 6$

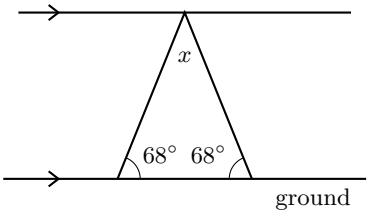
(b) $\angle 5$

(c) $\angle 8$

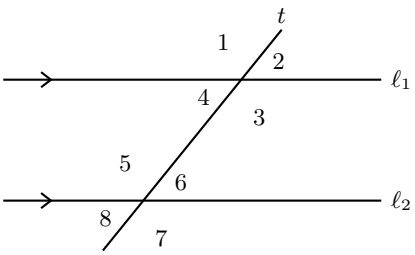
(d) $\angle 1$

(e) Suppose ℓ_1 and ℓ_2 were **not** parallel. Which of your answers would still be true? Explain.

11. The top bar of a swing set is parallel to the ground. Each leg makes a 68° angle with the ground. Find x, the angle between the two legs at the top. Explain your steps.



12. On a farm near Brooks, two straight irrigation ditches, ℓ_1 and ℓ_2, are parallel. A straight gravel road, t, crosses both ditches. The road makes $\angle 2 = 51°$ with the first ditch.

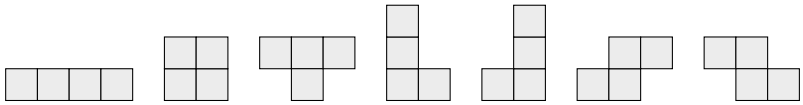


(a) Find $\angle 6$. Give a reason.

(b) Find $\angle 5$. Give a reason.

(c) Further along, the road crosses a third ditch. There, the angle in the same position as $\angle 2$ is 50°. Is the third ditch parallel to the first one? Explain.

13. A **tetromino** is a shape made of 4 equal squares joined edge to edge. Kim drew these shapes and says, "There are 7 different tetrominoes."



(a) Kim counted some shapes twice. Which shapes are really the same? Explain.

(b) How many different tetrominoes are there?

(c) What could Kim do to avoid this mistake next time?

Part B: Multiple Choice

14. Which angle is a reflex angle?

A. 245° **B.** 95° **C.** 180° **D.** 89°

15. The supplement of 64° is

A. 26° **B.** 116° **C.** 296° **D.** 154°

16. A boat heads south-east. Its true bearing is
A. 315° **B.** 225° **C.** 135° **D.** 045°

17. Line m crosses line n, making an angle of 89°. How should the pair be sorted?

A. Neither: they cross, but not at 90°.

B. Perpendicular: 89° is about 90°.

C. Parallel: they go almost the same way.

D. Both parallel and perpendicular.

18. In the diagram for Question 9, $\angle 2$ and $\angle 8$ are
A. corresponding **B.** alternate interior **C.** vertically opposite **D.** alternate exterior

19. $\ell_1 \parallel \ell_2$ and $\angle 1 = 109°$ (same numbering). Then $\angle 8$ is
A. 109° **B.** 71° **C.** 19° **D.** 251°

20. The sum of the angles in a polygon with 9 sides is
A. 1620° **B.** 1440° **C.** 1260° **D.** 1080°

21. The fewest moves to solve the Tower of Hanoi with 7 discs is
A. 49 **B.** 128 **C.** 63 **D.** 127

Part C: Numerical Response

22. Numerical Response At exactly 5:00, the smaller angle between the hands of a clock is _____°.

23. Numerical Response A tangram square has a total area of 144 cm^2. The area of one small triangle is _____ cm^2.

24. Numerical Response A canoe travels on a bearing of 326°. The bearing to paddle straight back is _____°.

25. Numerical Response An angle of 137° is bisected. Each new angle measures _____° (to the nearest tenth).
